PROJECT DEVELOPMENT PHASE

Cyber Security With AI – Model Performance Test

Date: 19 June 2026

Team ID: -----

Project Name: AI-Enhanced Intrusion Detection System

Maximum Marks: 10 Marks

Model Performance Testing

S.No.	Parameter	Values	Screenshot
1	Information Gathering	**Footprinting:** Identified network traffic characteristics, attack signatures, NSL-KDD dataset attributes, and system architecture. **Reconnaissance:** Collected information on attack patterns including DoS, Probe, R2L, and U2R attacks for training and testing purposes.	Attach Dataset Analysis Screenshot
2	Scanning the Target	**Scanning Info:** Network traffic samples were analyzed using the AI-based intrusion detection model. The system scanned input features and classified traffic into attack categories. **Risk**	Attach Prediction Dashboard Screenshot

S.No.	Parameter	Values	Screenshot
		Factors: Unauthorized access attempts, denial-of-service attacks, port scanning activities, and privilege escalation attempts were identified as major threats.	
3	Gaining Access	**Access Process:** Simulated attack traffic from the NSL-KDD dataset was processed through the intrusion detection engine to evaluate security vulnerabilities. **Vulnerability Found:** High-volume traffic anomalies, failed login attempts, suspicious scanning behavior, and compromised system indicators were detected successfully.	Attach Attack Detection Screenshot
4	Maintaining Access – Automation (AI Implementation)	**AI Tools Used:** Python, Flask, Scikit-learn, Random Forest Classifier, Pandas, NumPy. **Automation Implemented:** Automated threat detection, attack classification, confidence score generation, alert	Attach Automation Workflow Screenshot

S.No.	Parameter	Values	Screenshot
		logging, and dashboard monitoring were implemented using machine learning algorithms.	
5	Covering Tracks & Report	**Vulnerability Risk Factors:** DoS attacks, Probe attacks, Remote-to-Local attacks, User-to-Root privilege escalation attacks, and suspicious network behavior. **VAPT Report:** The AI model achieved approximately 96.8% accuracy in identifying malicious network activities. Real-time monitoring and alert generation improve cybersecurity response and reduce potential security risks.	Attach Final VAPT Report Screenshot

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Cyber Security Assessment Summary

Threat Categories Detected

1. Denial of Service (DoS)
2. Probe / Port Scan Attacks
3. Remote to Local (R2L) Attacks
4. User to Root (U2R) Attacks
5. Normal Network Traffic

Security Risk Analysis

Risk Type	Severity	Detection Status
DoS Attack	High	Detected
Probe Attack	Medium	Detected
R2L Attack	High	Detected
U2R Attack	Critical	Detected
Suspicious Traffic	Medium	Detected

AI-Based Security Automation

Implemented Features

- Automated Intrusion Detection
- Machine Learning-Based Classification
- Real-Time Threat Monitoring
- Confidence Score Prediction
- Alert Logging and Reporting
- REST API-Based Detection Service

AI Technologies Used

- Random Forest Classifier
- Scikit-learn
- Python
- Flask
- Pandas
- NumPy
- StandardScaler

Vulnerability Assessment Report

Findings

- Network attacks can be detected automatically using machine learning.
- Real-time classification reduces response time to security incidents.
- Automated alerts improve monitoring efficiency.
- The system effectively distinguishes between normal and malicious traffic.

Recommendations

- Regularly retrain the model using updated threat datasets.
- Integrate cloud-based monitoring for enterprise deployment.
- Implement advanced threat intelligence feeds for emerging attacks.
- Perform periodic security audits and penetration testing.

Conclusion

The AI-Enhanced Intrusion Detection System successfully combines Cyber Security and Artificial Intelligence to identify and classify network threats in real time. The implementation of machine learning-based automation improves threat detection accuracy, minimizes manual intervention, and enhances overall network security. The project demonstrates the practical use of AI for vulnerability assessment, intrusion detection, and cybersecurity monitoring.